CrewSight

Live crew visibility & fleet intelligence for rowing regattas, training, and broadcast.

CrewSight connects on-water Android devices to Altitude HD’s cloud platform — delivering real-time positions, hull motion events, zone awareness, and production-ready overlays for race operations and live streaming.

Technical overview · Altitude HD · 14 June 2026

Platform: **AHD - LiveStream** · ahd-livestream.vercel.app

What CrewSight delivers

On-water visibility

- Live GPS positions for every equipped crew
- Speed, heading, and recent track trails
- Configurable reporting intervals (1 s – 30 s+)
- Dual data sources: CrewSight recorder or Traccar trackers

Operations intelligence

- Fleet maps with club branding and device search
- Geofenced zones — course, warm-up, hazard, marshal areas
- Zone-entry alerts and fleet statistics bar
- Session history and route replay

Hull & crew event detection

- Accelerometer-based hull event signals from the device
- Capsize / SOS device alarms surfaced on the map
- Sudden-stop detection after recent rowing speed
- Audible ops alert with acknowledge workflow

Broadcast & regatta integration

- Transparent vMix browser overlays (1920×1080)
- RowIT CSV feeds — draw, results, progression, logos
- CV leader-line overlay from drone computer vision
- Regatta dashboards for flat-water and beach sprints

System architecture

```
CrewSight GPS (Android)          Traccar GPS devices (optional)
        │  HTTPS POST                        │  API poll
        ▼                                    ▼
   Recorder ingest API  ◄───────────►  AHD - LiveStream (Vercel)
   (Neon Postgres)                     ├── Fleet maps & zone alerts
        │                              ├── Operations monitor
        │                              ├── vMix overlays & race graphics
        ▼                              └── CV position API (Vercel KV)
   Session storage · history · device registry
```

CrewSight devices upload JSON samples to the recorder ingest endpoint. The AHD - LiveStream hub polls live snapshots for maps, overlays, and alerts. Operators choose **Recorder (CrewSight)** or **Traccar** as the position source per deployment.

CrewSight GPS — mobile app

Native Android app for regatta and training sessions on supplied GPS handsets.

Capability	Detail
GPS tracking	Latitude, longitude, speed, heading, fix time; configurable interval
Motion sensors	Accelerometer — tilt and hull motion for event detection
Device telemetry	Battery %, session heartbeat, device ID
Upload path	HTTPS batching (~3 s flush) to cloud ingest API
Typical ingest rate	~0.5 Hz at 30 s GPS · up to 1 Hz for high-rate modes (M26 field-validated @ 1 Hz, Jun 2026)
Field endurance	~6–8 h+ per charge (device-dependent, typical regatta day)

Install Side-loaded APK · ingest URL configured once · “Test upload connection” before going afloat

Cloud platform — AHD - LiveStream

Fleet maps

- **Altitude HD fleet map** — live devices, route history, custom markers, speed screen
- **Club / venue maps** — branded fleet views (e.g. Karāpiro, RNZ RowSafe-style layout)
- **Beach Sprints map** — coastal course, buoys, timing lines, head-to-head analysis
- Colour-coded speed trails · predict-to-now smoothing · map refresh rate control

Geofencing & zone alerts

- Import Traccar geofences or define course boundaries
- Zone types: course, warm-up line, hazard, marshal — colour-coded on map
- Devices outside designated zones trigger visual alerts and stats-bar counts
- Optional email notifications via Resend for zone events (configurable recipients)

Hull & crew event panel

Operations maps include a dedicated event panel for on-water incidents:

- Device-reported SOS / capsize alarm attributes
- Sudden stop after confirmed rowing speed (speed > 1.2 m/s, then < 0.35 m/s for 45 s)
- Map marker flash, fleet list highlight, audible tone for new unacknowledged events
- One-click acknowledge — clears until the next event on that device

Operations monitor

Web dashboard for race control and technical staff (recorder backend).

Function	Description
Live fleet map	All active devices with status and last fix age
Device list	Search, filter, ingest rate, battery level
Geofence editor	Define and manage course zones
Regatta messages	Broadcast text to device sessions
History API	Route replay, session export, Postgres-backed retention
Storage monitor	Database usage against configurable quota

Data captured per sample

Field group	Fields
Position	GPS lat/lng, speed, heading, accuracy, fix timestamp
Motion	Accelerometer axes, tilt, stroke/motion indicators (mode-dependent)
Events	Hull event flags, device alarm attributes, capsize signal
Device	Device ID, session ID, battery %, heartbeat
Optional (RNZ recorder)	Heart rate where enabled on full recorder app

At 30 s GPS, expect ~0.5 Hz ingest and ~43k samples/device/24 h. Cellular data per device is typically a few MB per session.

Broadcast & production tools

vMix overlays (transparent 1920×1080)

- Lower thirds, draw boards, results, leader lanes, live tracking graphics
- Keyboard triggers from hub or direct on overlay pages
- RowIT regatta code drives CSV data — club logos from AHD_lookup
- Weather map, speed charts, and venue-specific layouts (Milford, Karāpiro, beach sprints)

CV leader line

Computer-vision position from the drone laptop posts to `/api/cv-position` (Vercel KV). vMix polls the transparent leader overlay — same streamId as GPS / live production. TouchDesigner OSC path remains available locally.

Regatta dashboards

- Flat-water regatta dashboard — RowIT progression, heat results, club logos
- Beach Sprints regatta dashboard — knockout trees, CSV archive fallback
- Daily CSV snapshots archived locally if RowIT links go offline

Connectivity & deployment

Component	Recommendation
Handset	One NZ Smart **M26** (4G Android) — pre-configured for CrewSight; 5,000 mAh
Cellular	Managed IoT SIM — regatta-day activation · shared APN with public HTTPS
Cloud	Vercel (AHD - LiveStream) + Neon Postgres (recorder) + Vercel KV (CV positions)
Handset settings	Location "Allow all the time" · battery unrestricted · notifications on

Reporting rate profiles

CrewSight supports flexible upload intervals to balance visibility, battery, and data cost:

Mode	Interval	Typical use
High visibility	1 s	Start sequences, finals, incident monitoring
Active racing	5 s	Course tracking during heats and semis
Standard	30 s	Training, warm-up, long sessions — best battery life

Field validation — One NZ Smart M26 (14 Jun 2026)

Test	M26 (H7)	S21 reference (H6)
GPS fix rate (~10 km drive)	1 Hz	1 Hz
Median accuracy	**3.3 m**	5.2 m
p90 accuracy	**5.2 m**	7.6 m
Max fix gap	1 s	2 s
Track agreement	Median 4.2 m separation (100% matched ±3 s)	
Battery (6.5 h @ 1 Hz)	96% → 76% · ~3.1 %/h · ~32 h from 100%	
8 h regatta day @ 1 Hz (est.)	~25% drain — overnight charge sufficient	

Why CrewSight

- **End-to-end** — device app, cloud ingest, ops maps, and broadcast overlays from one team
- **Regatta-ready** — RowIT integration, club logos, multi-venue support
- **Scalable** — tested architecture to 270+ concurrent devices
- **Production-aware** — built by Altitude HD for live streaming and race operations

Important. CrewSight is a situational-awareness and fleet-visibility platform for rowing operations and broadcast. It is not certified rescue equipment and does not replace official race rules, patrol boats, qualified officials, or personal flotation requirements. Event detection features assist operators in identifying possible on-water incidents — response decisions remain with event organisers and qualified personnel.